Suraj Kumar Choudhary

HPC-AI Engineer | System Administrator | AI/ML Developer

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PROFESSIONAL SUMMARY

Results-driven HPC-AI Engineer with hands-on experience managing India's fastest supercomputers (Airawat and PARAM Siddhi AI) at C-DAC ACTS Pune, contributing to systems with 17.5 PF peak compute capacity. Skilled in HPC cluster administration, SLURM workload management, Linux systems, AI/ML model development, and full-stack web development. Passionate about applying machine learning to predictive maintenance, anomaly detection, and high-performance computing optimization.

WORK EXPERIENCE

HPC-AI Engineer (Project Engineer) | **C-DAC ACTS Pune** *Mar 2025 – Present*

National Supercomputing Mission (NSM) | NPSF Team

- Manage and maintain India's fastest supercomputers — Airawat and PARAM Siddhi AI — ensuring stable operation at 17.5 PF peak compute capacity.
- Administer SLURM workload manager for efficient job scheduling, resource allocation, and queue management across large-scale HPC clusters.
- Monitor cluster health using Prometheus and Grafana dashboards; proactively detect and resolve node/network failures to maintain high availability.
- Contributing to the SDDG PoC (Secure Distributed Data Grid) project under NSM, enabling high-speed 400G data transfer between HPC sites (Pune–Bangalore) for secure job migration and resource utilization optimization.
- Perform routine system administration tasks: OS patching, firmware updates, storage management, and user account provisioning on Linux-based HPC nodes.

AI/ML Intern | **NIELIT (Work-Based Learning)** *Feb 2024 – Jul 2024*

- Developed and trained machine learning models for classification and prediction tasks using Python, Scikit-learn, and TensorFlow.
- Conducted data preprocessing, feature engineering, and model evaluation to improve accuracy and generalization on real-world datasets.
- Gained practical exposure to deep learning architectures, computer vision techniques, and model deployment workflows.

Front-End Developer Intern | **LadyBird Web Solution Pvt. Ltd.** *Jun 2023 – Aug 2023*

- Designed and developed responsive web interfaces using React.js, JavaScript, HTML5, and CSS3.
- Collaborated with backend teams to integrate RESTful APIs and deliver dynamic user-facing features within Agile sprint cycles.
- Improved UI performance and cross-browser compatibility; contributed to component library development.

PROJECTS

SDDG POC — Secure Distributed Data Grid (NSM) · *HPC · Networking · Linux · 400G Interconnect*

- Leading a Proof-of-Concept for high-speed secure data transfer between HPC clusters (Pune–Bangalore) under the National Supercomputing Mission.
- Focused on enabling efficient job migration, secure multi-site data movement, and optimized resource utilization across geographically distributed clusters at 400 Gbps throughput.

HPC Node/Cluster Failure Prediction with Prevention · *Python · Grafana · ML · SLURM* [GitHub]

- Designed and implemented an AI-powered predictive maintenance system analyzing real-time HPC metrics (CPU, memory, latency, disk I/O) from Prometheus and Grafana.
- Trained ML models for anomaly detection and failure prediction, significantly reducing unplanned downtime and improving cluster reliability.
- Automated alerting and prevention workflows via Shell scripts integrated with SLURM job scheduler.

Vehicle Detection & Counting System · *Python · YOLOv5 · Deep SORT · OpenCV · Streamlit* [GitHub]

- Built a real-time computer vision application to detect, classify, track, and count vehicles from video feeds using YOLOv5 for object detection and Deep SORT for multi-object tracking.
- Developed an interactive Streamlit dashboard for visualization and reporting of vehicle analytics.

HPC Infrastructure Provisioning xCat · *xCAT · Rocky-Linux · Unified Dashboard* [GitHub]

- xCAT cluster management, automating server discovery, PXE booting, OS installation, and provisioning through unified management interfaces.

Stock Price Predictor Web App · *Python · Machine Learning · Streamlit · Jupyter* [GitHub]

- Built an ML-powered stock price prediction web application integrating real-time market data with predictive models to deliver actionable trading insights.

IPL Match Win Probability Predictor · *Python · ML · Streamlit · Jupyter* [GitHub]

- Developed a Streamlit web app using machine learning to predict IPL match win probabilities based on live inputs: teams, current score, overs, and wickets.

GPT 1B Model Training From Scratch · *Nvidia-GPU-A100 · Model-Training* · [GitHub]

- A complete 1B-parameter GPT implementation from scratch using PyTorch, DistributedDataParallel, Gradient Checkpointing, and 8× NVIDIA A100 GPUs.

TECHNICAL SKILLS

HPC & Systems: SLURM, PBS/Torque, HPC Cluster Administration, Linux (RHEL/CentOS/Ubuntu), Bash Scripting, System Administration

AI / ML: Python, Scikit-learn, TensorFlow, Keras, YOLOv5, Deep SORT, OpenCV, Anomaly Detection, Predictive Modeling

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3), Prometheus, Grafana, CI/CD

Web Development: React.js, JavaScript (ES6+), HTML5, CSS3, Streamlit, REST APIs

Networking & Storage: High-Speed Interconnects (400G), InfiniBand, Parallel File Systems, Data Grid Computing

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, Zabbix

EDUCATION

PG Diploma in HPC System Administration (PG-DHPCSA) *Aug 2024 – Mar 2025*
C-DAC ACTS Pune, Maharashtra

Bachelor of Engineering (B.E.) *Jul 2020 – Aug 2024*
Mahant Bachittar Singh College of Engineering and Technology, Jammu

CERTIFICATIONS & TRAINING

- PG-DHPCSA — C-DAC ACTS Pune (HPC System Administration, SLURM, Linux, Parallel Computing)
- Work-Based Learning (WBL) in AI/ML — NIELIT, Jammu
- Front-End Web Development — LadyBird Web Solution Pvt. Ltd.

LANGUAGES

English — Professional Working Proficiency | **Hindi** — Native / Bilingual Proficiency